Safety and Health at Work 15 (2024) 379–381

Contents lists available at ScienceDirect

Safety and Health at Work

journal homepage: www.e-shaw.net

Commentary

Applications and Impact of Industry 4.0: Technological Innovations in Occupational Safety and Health

David Koh [1], Alvin Tan [2,*]

[1] Saw Swee Hock School of Public Health, National University of Singapore, 12 Science Drive 2, #10-01, Singapore 117549, Singapore
[2] Ministry of Health Holdings, 1 N Buona Vista Link, #09-01 Elementum, Singapore 139691, Singapore

ARTICLE INFO

Article history:
Received 6 May 2024
Received in revised form
26 June 2024
Accepted 2 September 2024
Available online 7 September 2024

The future of work is changing faster than ever with the rapid growth and adoption of technological innovations, heralding the emergence of a new Occupational Safety and Health (OSH) paradigm [1]. The acceleration of technological innovations in the field of OSH is driven by a confluence of factors. First, technology is everywhere and has evolved to become a defining characteristic and presence in nearly every facet of our daily lives. This has led to an exponential growth of data ("tsunami of data"), fueling advancements in data analytics, machine learning, artificial intelligence (AI), and robotics, creating synergistic opportunities for technological innovations. Second, more jobs and tasks are also increasingly vulnerable to automation with the adoption of technology that promises increased efficiency and productivity at a much lower labor cost. Third, the nature of career is changing as society shifts away from the traditional functional model of a single, long-lasting linear progressing career to the normalization of intentional job-hopping driven by opportunism. There is now a generational change and diversification of the workforce with the influx of digitally native and tech-savvy younger workers adept in digital technologies, fostering a culture that is open to technological changes and innovations. The confluence of these factors has resulted in a synergistic acceleration and advancement of technological innovations.

The opportunities and advancements in OSH with the adoption of these technological innovations offer solutions to address enduring issues in traditionally known OSH-related risks, especially ergonomics and safety-related risks [2]. Technological innovations that had reshaped the field of OSH in 2023 [3] include the following:

- Wearable Technology. Wearable technology, such as phase change cooling vests [4], biometric smart wear with health status and environmental sensors, and personalized exoskeletons [5,6], provides accurate real-time feedback and personalized risk assessment data such as physical strain, ergonomics of work, environmental exposure levels, and fatigue levels [7]. Wearables empower workers to make informed decisions when exposed to workplace risks, enhance workplace performance, and facilitate companies to monitor worker safety and optimize safety protocols.
- Drones. Drones are utilized for safety inspections in hazardous environments, accessing remote or dangerous areas, inspecting infrastructure, and monitoring compliance [8]. They allow for improved efficiency while excluding the worker and human factors in the performance of tasks that typically pose health and safety risks to the workers [9].

David Koh: https://orcid.org/0000-0001-6803-7879; Alvin Tan: https://orcid.org/0000-0002-2302-8247
* Corresponding author. Ministry of Health Holdings, 1 N Buona Vista Link, #09-01 Elementum, Singapore 139691, Singapore.
E-mail address: alvin.tankw@mohh.com.sg (A. Tan).

2093-7911/$ – see front matter © 2024 Occupational Safety and Health Research Institute. Published by Elsevier B.V. on behalf of Institute, Occupational Safety and Health Research Institute, Korea Occupational Safety and Health Agency. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).
https://doi.org/10.1016/j.shaw.2024.09.001

- Collaborative Robots. Also known as Cobots, they work alongside the workers, assisting with repetitive tasks and heavy lifting, removing worker's contact with hazardous applications, and incorporating additional safety features such as force-limiting technology and collision detection [10]. The integration of Cobots into the work processes enhances productivity while minimizing injury rates.
- Gamification and Immersive Technologies (e.g., virtual reality and augmented reality). Gamification incorporated into the training of workers introduces new methodologies to deliver safety training and awareness [9]. Through virtual or augmented reality and gamified modules with immersive life-like interactive simulations of hazardous situations, gamification offers realistic and engaging training and education on safety practices [11,12].
- AI. AI-driven information-gathering safety systems allow for the analysis and processing of historical incident data, sense-making, and identification of trends in workplace surveillance and monitoring [13]. AI offers the ability to quickly consolidate, de-duplicate, and sense-make large amounts of information and surveillance data over traditional surveillance methods. With generative AI on the horizon, AI-driven safety systems have the potential to predict risks and assist in decision-making, resource allocation, and emergency response, enhancing overall resilience [14].
- Industrial Internet of Things (IoT). Industrial IoT connects machinery, sensors, and devices within industrial settings that provide real-time monitoring of safety risks, predictive maintenance, and early fault detection. The move towards digital management of operations and wireless communication, which can be done remotely, offers an interconnected data management that can improve safety and operational efficiency [1,2].
- Workplace Safety Reporting Software. Digital platforms help to streamline incident reporting, investigation, and corrective actions, with automated workflows that can keep track and ensure timely responses and compliance with regulations [15].

The Gartner Hype Cycle [16] offers a useful framework for allowing companies to understand the common adoption pathway of these technological innovations and to make informed decisions. The Gartner Hype Cycle describes 5 phases: Technology Trigger, Peak Of Inflated Expectations, Trough Of Disillusionment, Slope of Enlightenment, and Plateau of Productivity (Table 1), providing a graphical representation of the general trends of technological innovations, state of maturity, and future progression.

Table 1
Description of the 5 phases of the Gartner Hype Cycle

Phase	Description
Technology Trigger	The start of the Gartner Hype Cycle where the technology is newly introduced, sparking initial excitement with the potential of the technology.
Peak of Inflated Expectations	Enthusiasm for the technology reaches a zenith, potentially exceeding realistic capabilities
Trough of Disillusionment	Challenges in implementation and lack of demonstrable results of the technology becoming apparent, leading to disappointment and waning of the initial excitement.
Slope of Enlightenment	Practical applications of the technology emerge, use cases defined, and best practices for successful implementation are established
Plateau of Productivity	The technology achieves widespread adoption and becomes a well-integrated component of OSH practices, and value proposition is demonstrably positive

More importantly, beyond using the Gartner Hype Cycle to make future predictions and identify potential barriers to adoption, companies should adopt a multidisciplinary approach and collaboration between engineers, data scientists, researchers, OSH professionals, management, and workers to address the multifaceted challenges and specific needs with the integration of technological innovations into OSH practices, ensuring that the promise of technological innovations aligns with eventual practical and sustainable OSH outcomes.

With the integration of technological innovations, new and emerging risks will be introduced and concerted efforts must be made to control and manage the following new and emerging risks [2,17].

- Impact on Mental Health. The ease of digital OSH surveillance and monitoring with real-time feedback supported by technological innovations can lead to pervasive monitoring. The workers may experience a loss of autonomy and control over their work, invasion of personal privacy, and feelings of job insecurity and displacement as they are constantly being monitored for lapses that can result in penalties. Such pervasive monitoring can cause increased pressure, stress, and anxiety for the workers.
- Technostress. Workers who are less tech-savvy may not be able to cope with the pressure to constantly adjust to new technologies, especially if proper training processes are not in place. Workers who are unable to utilize the technologies in a healthy and productive manner may suffer from stress and cognitive overload.
- Physical and Safety Hazards. The complexity of the systems and cobots may bring about unexpected reactions or unanticipated patterns that the workers cannot predict. Given the closer collaborations, this may potentially increase the risk of accidents and worker–robot collisions as the worker will not be able to react in a timely manner. In addition, the human–machine interface can introduce new forms of repetitive tasks for workers that can cause strain injuries and musculoskeletal disorders.
- Cybersecurity Risks and Threats. The increased dependency on IoT devices and increased digital connectivity introduce the risk of data breaches with more entry points for cyberattacks. Without robust cybersecurity measures, sensitive company data and workers' information may be at risk, and there could be potential disruptions in safety-critical systems arising from system vulnerabilities.

Shifts in mindsets and behavior are necessary and critical for successful translation, implementation, and incorporation of technological innovations into lasting OSH solutions. Individual workers will need to embrace the inherent transformative impact and be familiar with technological innovations. This will be especially pertinent in a multigenerational workforce that will require greater collaboration and engagement to effectively support mature and less technologically adept workers to upskill and adapt to the technical aspects of technological innovations. Companies must be able to look beyond business success and contend with the need to raise productivity and reduce costs, which may be challenging. They will need to seize the opportunity to ride the technological wave and invest in harnessing disruptive technologies in the field of OSH to achieve safer processes, practices, and a safer workplace.

Other barriers in the implementation of technology-driven decision-making processes in OSH must be addressed. For example, regarding particular concerns of generative AI applications in OSH, issues such as transparency, data quality, bias and its mitigation, privacy and data security, human oversight, training and education, AI hallucinations, misinformation, liability and accountability, as well as ethics and regulatory compliance must be considered and addressed. Policymakers, ethicists, and regulators must catch up in tandem with the advancements to establish regulatory OSH frameworks and ethical guidelines [18].

While these regulations ensure responsible and ethical use of technological innovations, a balanced approach without laying out excessively stringent regulations that can hinder the development and implementation of technological innovations must be followed.

Technological innovations will continue to accelerate at an unprecedented rate, and we can expect an emerging era of evolving and transformative AI trends impacting companies and workplaces. Examples include the introduction of intelligent applications that can respond autonomously and dynamically, metaverse technologies to create unique experiences and impactful interactions at the workplace, development of an augmented-connected workforce, and establishment of AI Trust, Risk and Security Management, and Continuous Threat Exposure Management [19,20]. Correspondingly, these will influence and redefine the landscape of OSH as we adapt and navigate new OSH paradigms to create an overall safer and healthier workplace for workers.

Disclaimers

This paper is based on a keynote presentation entitled “Translating Knowledge into Action - Technological Innovations in OSH” which was delivered at the Asian Congress on Occupational Health held from November 22–25, 2023, in Daegu, Korea.

CRediT authorship contribution statement

David Koh: Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Alvin Tan:** Writing – review & editing, Writing – original draft, Conceptualization.

Conflicts of interest

We declare no conflict of interest for our submitted commentary, entitled “Applications and Impact of Industry 4.0: Technological Innovations in Occupational Safety and Health”.

References

[1] Akyildiz C. Integration of digitalization into occupational health and safety and its applicability: a literature review. Eur Res J 2023;9(6):1509–19.

[2] Arana-Landín G, Laskurain-Iturbe I, Iturrate M, Landeta-Manzano B. Assessing the influence of industry 4.0 technologies on occupational health and safety. Heliyon 2023;9(3):e13720.

[3] StartUs Insights. Top 8 workplace health and safety trends & innovations in 2023 [Internet]. 2023 [cited 2024 Apr 25]. Available from: https://www.startus-insights.com/innovators-guide/workplace-health-safety-trends/.

[4] Hu C, Wang Z, Bo R, Li C, Meng X. Effect of the cooling clothing integrating with phase change material on the thermal comfort of healthcare workers with personal protective equipment during the COVID-19. Case Stud Thermal Eng 2023;42:102725.

[5] EksoWorks. EksoWorks & ford partnership [Internet]. 2017 Nov 8 [cited 2024 Apr 25]. Available from: https://www.youtube.com/watch?v=HpYhY5-cVyk&t=22s.

[6] MākuSafe. MakuSafe wearable safety tech [Internet]. 2022 Apr 21 [cited 2024 Apr 25]. Available from: https://www.youtube.com/watch?v=SCBJdyolxhg.

[7] Svertoka E, Saafi S, Rusu-Casandra A, Burget R, Marghescu I, Hosek J, et al. Wearables for industrial work safety: a survey. Sensors (Basel) 2021;21(11):3844.

[8] GE Reports. Drones and high-powered AI technology are transforming safety inspections [Internet]. 2018 May 09 [cited 2024 Apr 25]. Available from: https://www.youtube.com/watch?v=UaEC5vLtEZw.

[9] Flor-Unda O, Fuentes M, Dávila D, Rivera M, Llano G, Izurieta C, et al. Innovative technologies for occupational health and safety: a scoping review. Safety 2023;9(2):35.

[10] Pauliková A, Gyurák Babeľová Z, Ubárová M. Analysis of the impact of human-cobot collaborative manufacturing implementation on the occupational health and safety and the quality requirements. Int J Environ Res Public Health 2021;18(4):1927.

[11] At Heights. 3M fall protection virtual reality experience for education & training [Internet]. 2018 Oct 02 [cited 2024 Apr 25]. Available from: https://www.youtube.com/watch?v=MKNPIlvMSGU.

[12] SAPJAPAN. The future of field service operations brought to you by SAP and Vuzix [Internet]. 2014 Mar 10 [cited 2024 Apr 25]. Available from: https://www.youtube.com/watch?v=5g7WUOB1uNk.

[13] Microsoft. Build 2017: workplace safety demonstration [Internet] [cited 2024 Apr 25]. Available from: https://www.youtube.com/watch?v=pL-c00M2CnI 2017 May 10; 2017 May 10.

[14] Shah IA, Mishra S. Artificial intelligence in advancing occupational health and safety: an encapsulation of developments. J Occup Health 2024;66(1):uiad017.

[15] Protex AI. Workplace AI safety software: protex AI overview [Internet]. 2023 Sep 26 [cited 2024 Apr 25]. Available from: https://www.youtube.com/watch?v=70AbYxyfC0s.

[16] Gartner. Gartner Hype Cycle [Internet]. 2022 [cited 2024 Apr 25]. Available from: https://www.gartner.com/en/research/methodologies/gartner-hype-cycle.

[17] Leso V, Fontana L, Iavicoli I. The occupational health and safety dimension of industry 4.0. Med Lav 2018;110:327–38.

[18] Koh D. Addressing concerns of Artificial Intelligence use in occupational health. J UOEH 2024;46(1):134.

[19] McCartney A. Gartner top 10 strategic technology trends for 2024 [Internet]. 2023 Oct 16 [cited 2024 Apr 25]. Available from: https://www.gartner.com/en/articles/gartner-top-10-strategic-technology-trends-for-2024.

[20] Hurst A. The top trends to impact tech providers by 2025 — Gartner [Internet]. 2023 Feb 01 [cited 2024 Apr 25]. Available from: https://www.information-age.com/top-trends-to-impact-tech-providers-by-2025-gartner-123501418/.